Chuanguo Xiong

Optical Metrology | Experimental Mechanics | Mechanical Engineering

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RESEARCH PROFILE

Mechanical engineering background combining optical measurement, experimental mechanics, and computer vision. Research interests include digital image correlation (DIC), deflectometry, full-field measurement, thermomechanical characterization, and 3D computer vision.

EDUCATION

M.Sc. in Mechanical Engineering | 2021-2024

Huazhong University of Science and Technology
Thesis: Thermomechanical Characterization of Packaged Structures via Fused 3D DIC and Thermography.

B.Eng. in Aircraft Manufacturing Engineering | 2017-2021

Nanjing University of Aeronautics and Astronautics

PUBLICATIONS

Gao Y, **Xiong C**, Lv W, and Zhu F. In-Situ Warpage Measurement of Electronic Packages with Projected Speckle DIC and Deep Learning-Based ROI Identification. IEEE Transactions on Instrumentation and Measurement (2024). DOI

Xiong C, Gao Y, and Zhu F. Specular Surface Deformation Measurement Based on Projected-Speckle Deflectometry with Digital Image Correlation. Optics and Lasers in Engineering 170 (2023): 107776. DOI

Gao Y, **Xiong C**, Lei X, and Zhu F. Optimized Super-Resolution Promote Accuracy for Projection Speckle Three-dimensional Digital Image Correlation. Measurement Science and Technology 34(11) (2023): 115601. DOI

Xiong C, Zeng B, Huang Y, and Zhu F. A Non-Destructive Inspection Method for Electronic Packaging Reliability Incorporating Mechanical and Thermal Information. IEEE 24th Electronics Packaging Technology Conference (2022): 719-724. DOI

Xiong C, Zeng P, Lv W, and Zhu F. Design and Optimization of a Novel MEMS Tuning Fork Gyroscope Microstructure. Micromachines 13 (2022): 172. DOI

Research & engineering experience

SELECTED RESEARCH

3D DIC and thermography for thermomechanical characterization

Combined stereo DIC and thermography to capture deformation and thermal information in packaged structures under thermal loading. Conducted mechanical testing to study localized deformation and structural response. Applied image reconstruction and spatial alignment methods to improve speckle matching and full-field measurement.

Automated region-of-interest extraction for warpage measurement

Implemented attention-enhanced U-Net++ architectures for automated region-of-interest extraction from heterogeneous surfaces. Applied network pruning and deep supervision to improve image processing. Evaluated the DIC framework against Moiré interferometry.

Finite element analysis and design of MEMS gyroscopes

Modeled MEMS microstructure dynamics in Ansys using modal and harmonic response analyses. Used the Taguchi method and signal-to-noise analysis to evaluate structural parameters and vibration transmission.

INDUSTRY EXPERIENCE

Embedded Software Engineer | Mindray Bio-Medical Electronics

Medical Equipment R&D team; joined July 2024.
Developed embedded software for ARM Cortex-based medical mechatronic systems, motion control, multi-sensor communication, and synchronization using UART and CAN. Worked with U-Net++ and PyTorch for automated visual analysis.

TECHNICAL SKILLS

Measurement and vision: DIC, thermography, OpenCV, camera calibration, U-Net++.
Mechanics and design: Ansys, Abaqus, SolidWorks, mechanical testing.
Programming: C/C++, Python, MATLAB, PyTorch.

SELECTED HONORS

Outstanding Graduate (2024)
China National Scholarship (2023)
First-Class Academic Scholarship (2021)